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EDUCATION

Xi'an Jiaotong-Liverpool University (XJTLU)

BSc Biological Sciences | *Bachelor* Sep 2021 – Jul 2025

- Language: Proficient English (IELTS 6.5).
- Major Courses: Practical Bioinformatics, Cell Biology Techniques, Fundamentals of Virology, Biochemical Methods, Advanced Genetics, Cell Signaling, Biotechnology, Structure and Dynamics of Biomolecules, Pathogenic Microbiology, Proteins in Action.

The University of Edinburgh

MScR Integrative Biomedical Sciences | *Master* Sep 2025 – Present

- Language: Proficient English.
- Work for Heng-jia Liu's Lab, mainly carry out bioinformatics analysis.
- Expect to graduate by October 2026.

INTERNSHIP & RESEARCH EXPERIENCE

China Academy of Medical Sciences Institute of System Medicine **Jun 2023 - Present**

Jun 2023 - Jul 2023 **Intern at Zhou Lab under the supervision of Professor Zhuo Zhou**

Studied and worked on experiments such as PCR, DNA recombination, plasmid preparation, cell culture and transfection, SDS-Page and Western-Blotting.

Jan 2024 - Present **Intern at CGT Platform under the supervision of Professor Jianwei Wang**

Molecular mechanism and physiological phenotype of SARS-CoV-2 viral protein NSP2 manipulation of the host translation system

- Conducted experiments such as PCR, DNA recombination, plasmid preparation, cell culture and transfection, SDS-PAGE, and Western Blot. Also engaged in bioinformatics experiments like Bulk RNA-seq and ChIP-seq. Independently built a bioinformatics analysis platform, which involved setting up a network-attached storage server, assembling, installing, and configuring relevant computer equipment, deploying automatic backup and redundancy systems, and maintaining the related equipment and data.

Deciphering the Temporal and Spatial Dynamics of Macrophage Polarization and Immune Modulation in Response to Poly I:C Stimulation

- This project aims to investigate the polarization, immune modulation, and tissue-specific responses of macrophages to Poly I:C (a synthetic analog mimicking viral double-stranded RNA) using a combination of bulk RNA sequencing, spatial transcriptomics, and single-cell RNA sequencing. My contributions included experimental design, sample collection and processing, and data analysis such as differential gene expression, time series analysis, and pathway enrichment.

Development of artificial intelligence large language model workflows and related auxiliary tools

- This project aims to develop an automated workflow (AI Workflow) based on large language models to quickly capture the latest scientific research news, organize and judge possible future trends through agents, and save them locally or send them to users via email. It can also be used to process nucleic acids, protein sequence data, etc., to achieve a variety of automated workflows.

Xi'an Jiaotong-Liverpool University (XJTLU) **Sep 2024 – Jul 2025**

Final Year Project under the supervision of Professor Rong Rong.

Establishment of a Nectin-1-Expressing CHO-K1 Cell Line to Study Monkey B Virus Entry into Human

- Pathogen and Problem: Monkey B virus (BV) is an alphaherpesvirus that commonly infects macaques but is highly pathogenic in humans, causing severe neurological disease.
- Research Objective: The study aimed to develop a cell model to investigate the mechanism of BV entry into human cells, specifically the interaction between BV glycoproteins and the human nectin-1 receptor.
- Methodology: A stable cell line expressing human nectin-1 was successfully established by transducing Chinese hamster ovary (CHO-K1) cells with a lentiviral vector carrying the human nectin-1 gene.
- Key Findings: The expression of the nectin-1 receptor in the new cell line was robustly confirmed at both the mRNA and protein levels using RT-qPCR, immunofluorescence assay, and Western blot.
- Conclusion and Future Work: The established cell line is a vital tool for studying BV entry mechanisms. Future work will focus on elucidating the BV glycoprotein-human nectin-1 interaction to understand viral entry and identify potential antiviral targets.

The University of Edinburgh **Sep 2025 - Present**

Work at Heng-jia Liu's Lab under the supervision of assistant Professor Heng-jia Liu

Bioinformatic characterization of CD39 and CD38 expressing T cells in colorectal cancer at single-cell resolution

- Disease of Study: Pulmonary lymphangioleiomyomatosis (LAM), an incurable multisystem neoplasm primarily affecting women, leading to lung failure.
- Target Molecule: B7-H3 (CD276), an immune checkpoint molecule often overexpressed in cancers and associated with poor prognosis, with a debated role in immune regulation.
- Core Scientific Question: To understand how B7-H3 regulates myeloid cell diversity within the LAM tumor immune microenvironment.
- Research Materials and Basis: Unpublished in-house CITE-seq data from a preclinical LAM mouse model where B7-H3 was suppressed in the LAM tumor cells. Preliminary analysis already shows that B7-H3 inhibition reduces specific myeloid populations, including immunosuppressive Fn1+ macrophages and classical monocytes.
- Project Aim: To perform further bioinformatic analysis on this dataset to delineate the mechanism by which B7-H3 governs the composition and function of myeloid cells, thereby understanding its role in remodeling the immune microenvironment in LAM.

Expect to finish by January 2026

During this time, the lab's bioinformatics analysis server suffered an SSH blast attack, and I independently hardened the server, which has not had any cybersecurity incidents so far.

Xi'an Jiaotong-Liverpool University (XJTLU) **Jul 2022 - Jan 2024**

Jul 2022 - Jun 2023 **Student Lecturer | Windows from Beginner to Proficient**

- The course helps students from the simplest operations to proficiency in using the Microsoft Windows operating system, using Microsoft 365 cloud services, learning advanced configuration of Windows systems, simple troubleshooting, and simple Windows CMD and Windows PowerShell scripting

Jul 2022 - Jun 2023 **Student Lecturer | Introduction to Computer Hardware Equipment System and Maintenance**

- The course covers the installation of Windows and Linux operating systems, daily maintenance and troubleshooting of Windows systems, personal computer configuration design, and practical installation and troubleshooting of computer hardware. Addition ally, it includes deep personalization of Windows using simple decompilation tools, configuring the Windows pre-installation en vironment, understanding the Windows Assessment and Deployment Kit (ADK), factory-customized deployment of Windows d esktops, and designing unattended auto-response installation configurations.

Jul 2022 - Aug 2022, Jul 2023 - Aug 2023 **Intern at Student Admissions & Career Development Office**

- Help the school prepare and mail acceptance letters for new students.Track logistics information and assist in handling logistics exceptions and package content exceptions.

CAMPUS EXPERIENCE

Xi'an Jiaotong-Liverpool University (XJTLU) | Class Monitor **Sep 2021 - Jun 2022**

- Responsible for coordinating with the teacher of the career planning course and classmates, completing course notifications and student feedback, and providing guidance and counseling on classmates' career planning
- Organized class activities, successfully assisted in several class outings, and was responsible for writing related communication drafts

Xi'an Jiaotong-Liverpool University (XJTLU) | Class Buddy **Sep 2022 – Jun 2023**

- Become a part of an important support system for students in university, provide timely assistance and guidance for junior students to effectively solve problems of life and study and improve communication ability and responsibility awareness of students
- Promote the mutual assistance between students, guide students to plan the campus life reasonably and assist the Student Support System in XJTLU Assisted in organizing training for officers to help freshmen carry out volunteer activities.

Xi'an Jiaotong-Liverpool University (XJTLU) | GAOKAO Consultation Fair **2022-2024**

- Go to various high schools to carry out admissions publicity and publicity activities to help GAOKAO candidates understand the school, help them choose the university, and encourage suitable candidates to apply for XJTLU.
- Provide assistance to candidates when filling in their preferences and help candidates choose their majors.
- Admission-related assistance is provided to admitted students to help them register and prepare for what they need for university.

SOCIAL EXPERIENCE

- More than 31 thousand followers on the video platform Bilibili (statistics as of Oct 2025). | Science and technology blogger, mainly publishes all kinds of computer use, maintenance guides, tips, tutorials, etc.
- Taobao store Tiancheng Computer ASYS Technology | Main products: computer machine, computer customization, computer accessories, computer peripherals, weak electricity equipment, computer repair services, softer support services, PC first aid kits.
- Operating website, URL: https://asystech.cn
- Pinduoduo store Office Software Assistant | Main products: Office and Windows support services, PC first aid kits.

- **Start own business and set up a company named ASYS technology studios.**

Main business:

- PC and server assembling and repairing.
- PC and server configuration design.
- NAS assembling and configuring.
- Windows installation, configuration and troubleshooting.
- Windows Batch Script Writing.
- Network configuration, soft router configuration.
- Server Operations and Maintenance

Served for:

- China Academy of Medical Sciences Institute of System Medicine, Suzhou institute of systems medicine.
- Yangzhou University.
- Zhejiang University
- Shenzhen Toeda Electromechanical Equipment Co., Ltd..
- ZJU-UoE institute.
- Soochou University.
- Many individual users.

AWARDS

- Xi'an Jiaotong-Liverpool University 2022-2023 academic year qualified buddy.
- Xi’an Jiaotong-Liverpool University Excellent Student Ambassador.

语言：简体中文（中国）ZH-CN

陈 思瞳

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教育

西交利物浦大学 (XJTLU)

生物科学 理学学士 | *本科 学士学位* 2021年09月 – 2025 年06月

- 语言水平: 雅思 A类 6.5 .

主要课程：实用生物信息学、细胞生物学技术、病毒学基础、生化方法、先进遗传学、细胞信号传导、生物技术、生物分子结构与动力学、病原微生物学、蛋白质的作用。

爱丁堡大学（UoE）

研究型硕士 生物医学科学| *硕士* 2025年09月 –至今

- 语言：英语
- 在刘恒佳实验室工作，主要从事生物信息学分析。
- 预计于2026年10月毕业。

实习与研究经历

中国医学科学院系统医学研究所 **2023年6月至今**

2023年6月 - 2023年7月 **在周实验室实习 ，师从周卓教授**

研究并参与了PCR、DNA重组、质粒制备、细胞培养与转染、SDS-Page和西部印迹等实验。

2024年1月-至今 **在王建伟院士指导下，在CGT平台实习**

SARS-CoV-2病毒蛋白NSP2控宿主翻译系统的分子机制及生理表型

- 进行了PCR、DNA重组、质粒制备、细胞培养与转染、SDS-PAGE和Western Blot等实验。还参与了生物信息学实验，如体体RNA测序和ChIP测序。独立构建了生物信息学分析平台，包括搭建网络连接存储服务器、组装、安装和配置相关计算机设备，部署自动备份和冗余系统，以及维护相关设备和数据.

解析巨噬细胞极化与免疫调节在多离子I：C刺激下的时间和空间动态

- 本项目旨在通过结合体体RNA测序、空间转录组学和单细胞RNA测序，研究巨噬细胞对Poly I：C（一种模仿病毒双链RNA的合成类似物）的极化、免疫调节及组织特异性反应。我的贡献包括实验设计、样本采集与处理，以及数据分析，如基因表达差异、时间序列分析和通路富集。

人工智能大语言模型工作流及相关辅助工具的开发

- 本项目旨在开发一款基于大语言模型的自动化工作流（AI Workflow），来快速捕捉最新的科研新闻，并通过Agent整理、判断可能的未来趋势，并且保存到本地或者通过邮件发送给用户。也可以用于处理核酸、蛋白质序列数据等，实现多种自动化工作流程。

西交利物浦大学 (XJTLU) **2024年9月 – 2025年7月**

荣荣教授指导的毕业项目。

建立表达Nectin-1的CHO-K1细胞系，研究猴B病毒进入人体的情况

- 病原体与问题：猴B病毒（BV）是一种α疱疹病毒，常见于猕猴，但在人类中具有高度致病性，可引发严重的神经系统疾病。
- 研究目标：本研究旨在开发一个细胞模型，以探讨细菌性病毒进入人类细胞的机制，特别是细菌性阴道炎糖蛋白与人类内线素-1受体之间的相互作用。
- 方法论：通过将携带人类nectin-1基因的慢病毒载体转导中国仓鼠卵巢（CHO-K1）细胞，成功建立了表达人类nectin-1的稳定细胞系。
- 关键发现：通过RT-qPCR、免疫荧光分析和WB，在mRNA和蛋白质水平均有强烈证据确认了新细胞系中内线素-1受体的表达。
- 结论与未来工作：已建立的细胞系是研究细菌性阴道炎进入机制的重要工具。未来工作将重点阐明BV糖蛋白与人类凝聚素-1的相互作用，以理解病毒进入并识别潜在的抗病毒靶点。

爱丁堡大学（UoE） **2025年9月至今**

在刘恒佳实验室工作，由刘恒佳助理教授指导

在结直肠癌中表达T细胞的CD39和CD38，单细胞分辨率下的生物信息学表征

- 研究疾病：肺淋巴管平滑肌瘤病（LAM），这是一种无法治愈的多系统肿瘤，主要影响女性，导致肺功能衰竭。
- 目标分子：B7-H3（CD276），一种免疫检查点分子，常在癌症中过度表达，且与预后不佳相关，在免疫调控中的作用存在争议。
- 核心科学问题：理解B7-H3如何调控LAM肿瘤免疫微环境中的髓系细胞多样性。
- 研究材料与基础：来自临床前LAM小鼠模型的未发表内部CITE-seq数据，其中B7-H3在LAM肿瘤细胞中被抑制。初步分析已显示，B7-H3抑制会减少特定的髓系群体，包括免疫抑制性的Fn1+巨噬细胞和经典单核细胞。
- 项目目标：对该数据集进行进一步的生物信息学分析，阐明B7-H3调控髓系细胞组成和功能的机制，从而理解其在重塑LAM免疫微环境中的作用。

预计将在2026年8月完成

在此期间，实验室的生物信息分析服务器遭受了一次SSH爆破攻击，我独立完成了服务器的安全加固，加固完成后，此服务器直到目前为止尚有过任何网络安全事件。

西交利物浦大学 (XJTLU) **2022年7月 - 2024年1月**

2022年7月 - 2023年6月 | 选修课学生讲师 | 课程：Windows从 入门到精通

1. 该课程帮助学生从最简单的作到熟练使用Microsoft Windows作系统、Microsoft 365云服务、学习Windows系统的高级配置、简单的故障排除，以及简单的Windows CMD和Windows PowerShell脚本编写

2022年7月 - 2023年6月 | 选修课学生讲师 | 课程：计算机硬件设备系统与维护导论

- 课程内容涵盖Windows和Linux作系统的安装、Windows系统的日常维护与故障排除、个人电脑配置设计，以及计算机硬件的实际安装与故障排除。此外，还包括使用简单的反编译工具进行深度个性化Windows、配置Windows预安装环境、理解Windows评估与部署工具包（ADK）、Windows桌面的工厂定制部署，以及设计无人值守自动响应安装配置。

2022年7月 - 2022年8月，2023年7月 - 2023年8月 **学生招生与职业发展办公室实习**

- 帮助学校准备并邮寄新生的录取通知书。跟踪物流信息，协助处理物流异常和包裹内容异常。

校园经历

西交利物浦大学 (XJTLU) | 班长 **2021年9月 - 2022年6月**

- 负责与职业规划课程教师及同学协调，完成课程通知和学生反馈，并为同学提供指导和咨询。
- 组织班级活动，成功协助多次班级外出活动，并负责撰写相关的通讯稿。

西交利物浦大学(XJTLU) | Class Buddy **2022年9月 – 2023年6月**

- 成为大学学生重要支持体系的一部分，及时为高三学生提供帮助和指导，帮助他们有效解决生活和学习问题，提升学生的沟通能力和责任意识。
- 促进指导学生合理规划校园生活，并协助XJTLU Student Support System 组织校园培训，协助新生开展志愿活动。

西交利物浦大学 (XJTLU) | 高考咨询会 **2022-2024**

- 走访各所高中开展招生宣传活动，帮助 高考 考生了解学校， 帮助 他们选择大学，并鼓励合适的考生申请XJTLU。
- 在申请人填写志愿时提供帮助，并协助他们选择专业。
- 为被录取的学生提供与招生相关的帮助，帮助他们注册并为大学所需的课程做好准备。

社会经历

- 视频平台Bilibili拥有超过3.2万粉丝（截至2025年11月统计数据）。| 科技和知识区博主，主要发布各种计算机使用、维护指南、技巧、教程等。
- 淘宝店 ASYS科技 |主要产品：电脑机、电脑定制、电脑配件、电脑外设、弱电设备、电脑维修服务、软件支持服务、电脑急救套件。
- 运营网站，URL: https://asystech.cn
- 拼多多商店：办公软件小助手 |主要产品：Office和Windows支持服务，PC急救包。
- **创办自己的公司，成立一家 名为ASYS科技工作室的公司.**

主要业务:

- 电脑和服务器的组装与维修。
- PC和服务器配置设计。

- NAS 组装与配置。
- Windows 安装、配置与故障排除。
- Windows 批处理脚本编写。
- 网络配置，软路由器配置。
- 服务器运营与维护。
- 生物信息分析服务器运维、网络安全配置。

主要客户:

- 中国医学科学院系统医学研究所，苏州系统医学研究所
- 扬州大学
- 浙江大学
- 中国邮电大学
- 深圳通意达机电设备有限公司
- 浙江大学爱丁堡大学联合学院
- 苏州大学

奖

- 西交利物浦大学 2022-2023学年 Qualified Buddy
- 西交利物浦大学 优秀学生大使